



Probability Theory and Mathematical Statistics

STAT 401 Midterm Examination • Time Allowed: 90 Minutes

Student Full Name: _____ Student ID: _____
Section / Lab: _____ Date: August 28, 2026

Run L^AT_EX again to produce the table

1. Conceptual Probability and Limit Theorems (15 Points)

- (a) (5 points) Under the Lindeberg-Lévy Central Limit Theorem, which condition is strictly necessary for the sample average $\bar{X}_n$ of i.i.d. random variables to converge in distribution to $\mathcal{N}(0, 1)$ after standardization?
- A. The underlying distribution of X_i must be symmetric.
 - B. The underlying distribution must have a finite variance ($\sigma^2 < \infty$).**
 - C. The sample size n must exceed 10,000.
 - D. The random variables must be uniformly bounded on $[0, 1]$.

Solution: Correct Choice: B. The classical CLT requires only finite mean μ and finite positive variance $\sigma^2 < \infty$.

- (b) (5 points) In a standard normal distribution $Z \sim \mathcal{N}(0, 1)$, what is the two-tailed probability of observing $|Z| > 1.96$?
- A. 0.10 (10%)
 - B. 0.05 (5%)**
 - C. 0.025 (2.5%)
 - D. 0.01 (1%)

Solution: Correct Choice: B. $P(Z < -1.96) = 0.025$ and $P(Z > 1.96) = 0.025$, giving a combined two-tailed probability of 0.05.

- (c) (5 points) For a 2×2 contingency table with total sample size N , row marginals R_1, R_2 , and column marginals C_1, C_2 , the expected cell count under independence is:
- A. $E_{ij} = (R_i \times C_j)/N$**
 - B. $E_{ij} = (R_i + C_j)/2$
 - C. $E_{ij} = (R_i \times C_j)/N^2$
 - D. $E_{ij} = \sqrt{R_i C_j}$

Solution: Correct Choice: A. By the product rule of independent events, $P(A_i \cap B_j) = P(A_i)P(B_j) = (R_i/N)(C_j/N)$, so $E_{ij} = NP(A_i \cap B_j) = (R_i C_j)/N$.

2. Derivation of Sample Odds Ratio and Chi-Square Statistic (20 Points)

- (a) (10 points) Consider a 2×2 table with cell counts $n_{11} = 40$, $n_{12} = 10$, $n_{21} = 20$, $n_{22} = 30$. Compute the sample Odds Ratio ($\widehat{OR}$) and interpret the result.

Solution: The sample Odds Ratio is:

$$\widehat{OR} = \frac{n_{11}n_{22}}{n_{12}n_{21}} = \frac{40 \times 30}{10 \times 20} = \frac{1200}{200} = 6.0$$

Interpretation: The odds of the positive outcome in the exposure group are 6 times the odds in the control group.

- (b) (10 points) Calculate the expected frequency E_{11} and its contribution to the Pearson χ^2 statistic.

Solution: $R_1 = 50$, $C_1 = 60$, $N = 100$.

$$E_{11} = \frac{50 \times 60}{100} = 30$$

Contribution: $\frac{(O_{11} - E_{11})^2}{E_{11}} = \frac{(40 - 30)^2}{30} = \frac{100}{30} = 3.33$.